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DAVID CHENG

www.linkedin.com/in/davidcheng16

SKILLS

Languages	C++, C, Python, Java
Platforms	Linux, Raspberry Pi, ESP32/ESP8266, Arduino
Tools	GDB, QEMU, GoogleTest, Logic Analyzer, Oscilloscope
Software	Git/GitLab, ClearCase, CMake, MATLAB, Qt/QML, DDS, Polarion, Jira, Confluence

WORK EXPERIENCE

Raytheon (Tewksbury, MA) – Software Engineer II May 2026 – Present

- Implemented a **DDS backend messaging layer** to connect legacy C code with a modern GUI, defining **IDL** contracts, writing response callbacks, and integrating with a frontend test driver.
- Resolved defects in **C** and validated fixes on target hardware, writing **shell scripts** to automate build packaging for integration and regression testing.

CNH Industrial (Oak Brook, IL) – Embedded Software Engineer II July 2024 – May 2026

- Developed features for a precision-farming application using **C++, Qt, and QML** on a **Yocto-based embedded Linux**, processing operator inputs and displaying real-time guidance data using **MVVM**.
- Validated customer features through integration testing in **QEMU** and on the target hardware at test sites, using **GDB** to diagnose software issues.
- Improved operator setup and load times by 20% by offloading persistence operations to a **background thread**, ensuring critical events are handled and displayed to the UI without delay.
- Implemented real-time guidance-line visualization by integrating line-generation with the display API, using **a dirty flag and observer pattern** to update the UI only when guidance data changes.

CNH Industrial (Oak Brook, IL) – Embedded Software Engineer I July 2022 - July 2024

- Led the development of an automated field-turn planner in **C++**, using a **Dubins-Clothoid algorithm** to generate the shortest drivable paths within a field while satisfying vehicle driving constraints.
- Designed a dynamic turn planning solution for collision turns by dividing them into sub-turns for re-planning, resulting in a formal **IP submission**.
- Validated algorithm behavior by writing **GoogleTest** unit tests with **mocked dependencies**, expanding test coverage for newly developed software.

PROJECTS

GitHub Portfolio**:** https://daviecheng.github.io/

Vision-Based Game Automation Project Sept 2024 – Present

- Developed C++ capture software on **Raspberry Pi 4** using **libcamera** and **OpenCV** to locate and crop a game device screen, validated on **116 real captures** across varied lighting and camera angles.
- Identified in-game indicators, such as health bars, in cropped screen images by implementing **HSV color masking** and **pixel-run detection** to determine game state for future automation.
- Built a **ring-buffer-style frame recorder** to retain debugging images and limit storage usage by replacing the oldest captures when the fixed limit is reached.

EDUCATION

New York Institute of Technology Sept 2017 - May 2022

M.S. Electrical & Computer Engineering GPA: 3.96/4.00

B.S. Electrical & Computer Engineering GPA: 3.89/4.00

Degree Honors: *summa cum laude*, Presidential Honor List (2018-2021)